

SRI SATHYA SAI INSTITUTE OF HIGHER LEARNING

ANANTAPUR CAMPUS



PROJECT REPORT

SUBMITTED BY : GIRIJA D

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PAPER TITLE : Smart-Parking-Management-System

INSTRUCTOR: Ms. Cherukuthota Divyasree

Department of Mathematics and Computer Science

Sri Sathya Sai Institute of Higher Learning

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ACKNOWLEDGEMENTS

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Institute of Higher Learning, for her valuable guidance, encouragement,
and support during the course of this work.*

I. PROBLEM STATEMENT

Traditional parking management involves manually tracking vehicles, parking slots, reservations, and billing. This can make it difficult to maintain accurate records and identify available slots. The Smart Parking Management System provides a computerized solution to manage these activities efficiently. It helps in tracking vehicles, managing parking slots, handling reservations, and generating parking bills in an organized manner.

II. OVERVIEW

The Smart Parking Management System is a Java-based application designed to manage vehicle parking efficiently using Object-Oriented Programming concepts. The system supports parking, slot reservation, billing, vehicle removal, vehicle search, parking history, and analytics. It uses file handling to store parking data and implements concepts such as inheritance, abstraction, encapsulation, interfaces, collections, and exception handling. The system also provides role-based access for Admin and Customer, ensuring organized and efficient parking management.

III. OBJECTIVES

- To develop a Java-based parking management system using Object-Oriented Programming concepts.
- To efficiently manage vehicles and parking slots.
- To provide parking slot reservation and cancellation facilities.
- To implement billing and discount calculation for parked vehicles.
- To provide vehicle search and parking history for easy record management.
- To implement Admin and Customer login for organized access.
- To use file handling to store parking data.
- To provide parking analytics such as occupied, available, and reserved slots.

IV.CONCEPTS USED

- Classes and Objects
- Constructors
- Encapsulation
- Inheritance
- Abstraction
- Method Overriding
- Interfaces
- Polymorphism
- Arrays
- ArrayList / Collections
- Exception Handling
- Packages
- File Handling
- Loops and Conditional Statements
- Static Members

V. TECHNOLOGIES USED

- Java
- Java Swing – for developing the graphical user interface
- Java Collections Framework – for managing data
- Java I/O – for file handling and data storage

- JDK (Java Development Kit) – for compiling and running the application

VI. SYSTEM DESIGN

1. Overview

The Smart Parking Management System follows a modular and object-oriented design. The system separates the graphical user interface, business logic, data management, and exception handling into different packages.

Main Components:

- GUI Layer: Provides user interaction using Java Swing.
- Service Layer: Handles parking, billing, reservations, search, history, login, file handling, and analytics.
- Model Layer: Represents vehicles and parking slots.
- Interface Layer: Defines common operations using interfaces.
- Exception Layer: Handles parking-related errors.
- Main Layer: Starts the application.

2. Vehicle System

Vehicle (Abstract Class)

- Stores common vehicle details such as vehicle number, owner name, and entry time.
- Declares `getRate()` and `getType()` methods.
- Provides common functionality for all vehicle types.

Concepts: Abstraction, Encapsulation, Inheritance, and Method Overriding.

Derived Classes

The system includes the following vehicle classes:

- Car
- Bike
- Truck
- Bus
- Electric Vehicle

Each class overrides the `getRate()` and `getType()` methods to provide vehicle-specific information and parking rates.

3. Parking System

ParkingSlot

- Represents an individual parking slot.
- Maintains the slot number, occupied status, reserved status, and vehicle details.
- Supports parking, vehicle removal, reservation, and reservation cancellation.

ParkingManager

- Manages 25 parking slots.
- Handles vehicle parking and removal.
- Allows removal of a specific vehicle using vehicle number or owner name.
- Displays the current status of parking slots.
- Prevents duplicate parking of the same vehicle.

Concepts: Encapsulation, Arrays, Collections, and Exception Handling.

4. Billing System

BillingManager

- Calculates parking charges based on parking hours and vehicle rate.
- Applies discounts for different categories.
- Calculates the final payable amount.

Formula:

Final Amount = Parking Amount — Discount

5. Reservation System

ReservationManager

- Reserves available parking slots.
- Prevents reservation of occupied or already reserved slots.
- Supports cancellation of reservations.
- Records reservation activities in the parking history.

6. Data Storage

FileManager

- Stores active parking information in parking_data.txt.
- Uses Java file handling to write parking details into the file.

Concept: File Handling.

7. History Management

HistoryManager

- Records important parking activities such as vehicle parking, vehicle removal, slot reservation, and reservation cancellation.
- Uses ArrayList to maintain the history records.

Concept: Collections.

8. Analytics

AnalyticsManager

Provides basic parking statistics such as:

- Total number of vehicles
- Available parking slots
- Reserved parking slots
- Number of different vehicle types

9. Search System

SearchManager

The system allows vehicles to be searched using:

- Vehicle Number
- Owner Name

The corresponding vehicle details are displayed when a match is found.

10. Login System

LoginManager

- Provides separate login access for Admin and Customer.
- Validates username and password.
- Allows a maximum of three unsuccessful login attempts.

Concepts: Conditional Statements, Encapsulation, and Static Members.

11. Graphical User Interface

The system uses Java Swing to provide separate screens for different operations:

- LoginFrame
- AdminFrame
- CustomerFrame
- ParkingFrame
- ReservationFrame
- SearchFrame
- BillingFrame
- HistoryFrame

Concepts: Swing Components, Event Handling, and GUI Programming.

VII. SPECIAL FEATURES

- Object-oriented design for modularity and maintainability
- Supports multiple vehicle types with different parking rates
- Parking slot reservation and cancellation
- Automated billing with discount support
- File-based parking data storage
- Specific vehicle removal using vehicle number or owner name
- Quick vehicle search by number or owner name
- Parking analytics for vehicles, available slots, and reserved slots
- Transaction history logging for parking activities
- Role-based access for Admin and Customer
- User-friendly graphical interface using Java Swing
- Exception handling for parking-related errors

VIII. WORK FLOW

The Smart Parking Management System follows a systematic workflow to manage vehicles, parking slots, reservations, billing, and records. The main steps of the system are:

1. System Start

The application starts and displays the login screen.

2. User Login

The user logs in as either Admin or Customer using valid credentials.

3. Role-Based Access

Based on the login role, the corresponding Admin or Customer interface is displayed.

4. Vehicle Management

The user can enter vehicle details such as vehicle number, owner name, and vehicle type.

5. Parking Slot Management

The system checks available slots and assigns an empty slot to the vehicle.

6. Reservation Management

Users can reserve an available parking slot and cancel a reservation when required.

7. Vehicle Search and Removal

Vehicles can be searched using the vehicle number or owner name. Specific vehicles can also be removed from the parking area.

8. Billing

The system calculates the parking charge based on the vehicle type and parking duration. Applicable discounts are also considered.

9. History and Data Storage

Parking activities such as parking, removal, reservation, and cancellation are recorded. Parking data is stored using file handling.

10. Analytics

The system displays information about occupied slots, available slots, reserved slots, and vehicle types.

11. System Completion

After completing the required operations, the user can exit the application.

IX. COMPILATION AND EXECUTION

The Smart Parking Management System is developed using Java and Java Swing. The program is organized into different packages containing model classes, interfaces, services, exceptions, GUI classes, and the main class.

Before running the application, all Java source files are compiled using the Java Development Kit (JDK). The compiled program is then executed through the main class.

Compilation Command:

The following command is used to compile all the Java source files in the project:

```
javac model\*.java interfaces\*.java service\*.java exception\*.java gui\*.java main\*.java
```

After successful compilation, the application is executed using the main class:

Execution Command:

```
java main.ParkingSystem
```

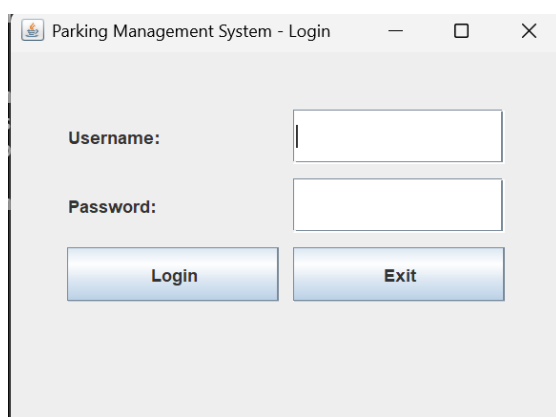
The compilation command compiles all the Java source files in their respective packages, while the execution command starts the Smart Parking Management System through the ParkingSystem main class.

X. CODE LINK:

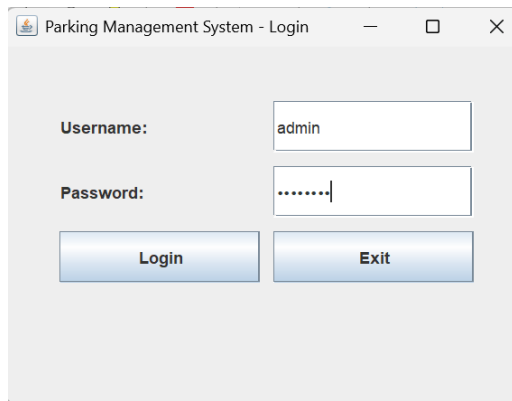
[Girijaaaa/Smart-Parking-Management-System: A GUI-based parking management system built with Java Swing for efficient vehicle and parking slot management.](#)

XI. SAMPLE RUN:

1.Login page:

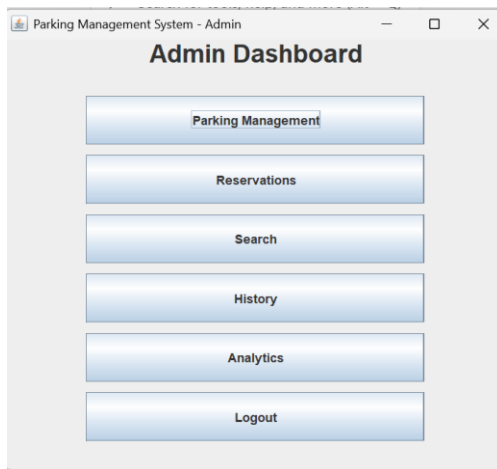


2.Login as admin:



The screenshot shows a window titled "Parking Management System - Login". It contains two input fields: "Username:" with the text "admin" and "Password:" with masked characters ".....". Below the fields are two buttons: "Login" and "Exit".

3.Admin Dashboard

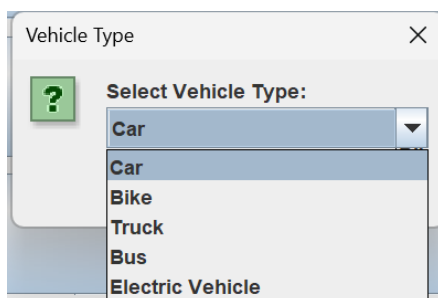
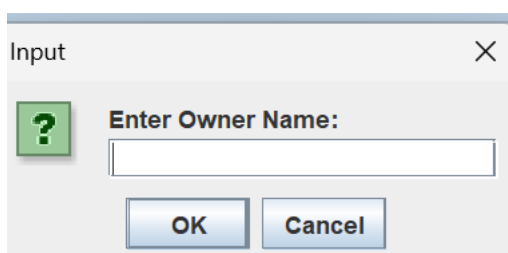
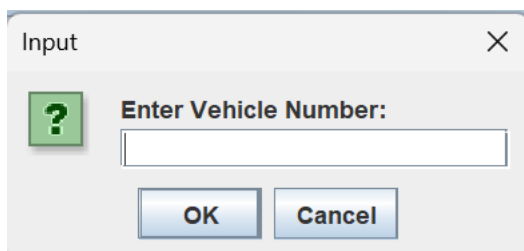


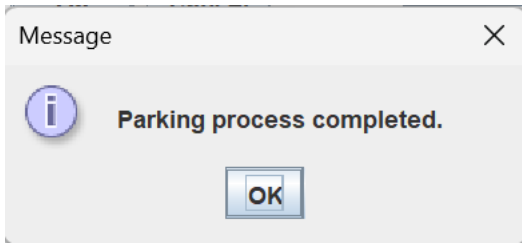
The screenshot shows a window titled "Parking Management System - Admin" with the subtitle "Admin Dashboard". It features a vertical stack of six buttons: "Parking Management", "Reservations", "Search", "History", "Analytics", and "Logout".

4.Parking Management Dashboard:

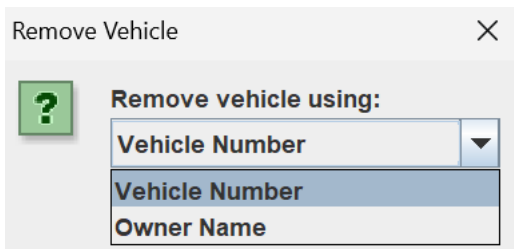


i.To park a vehicle:

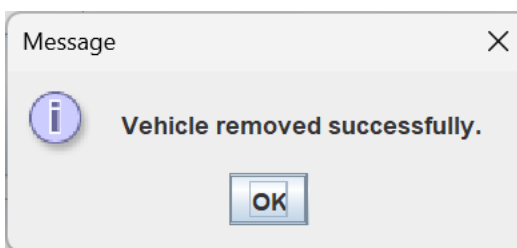
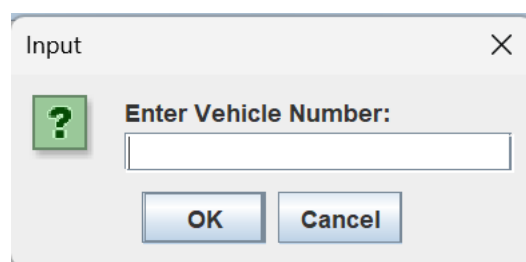
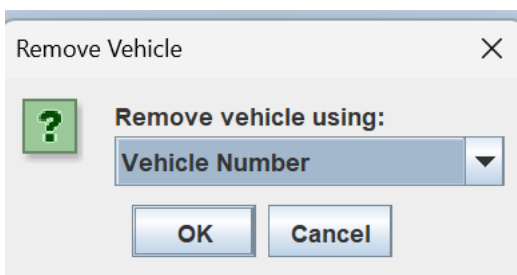




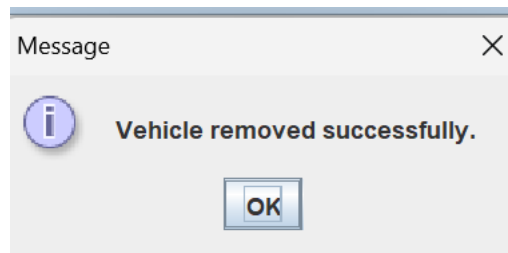
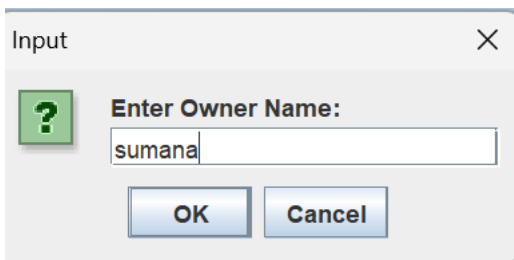
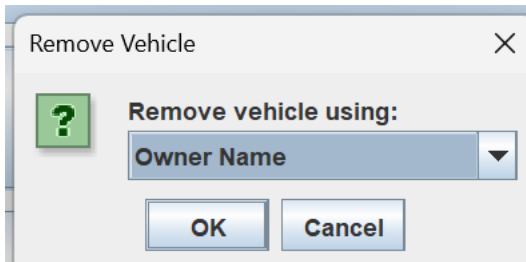
ii. To remove a parked vehicle:



a) Removing using vehicle number:



b) Removing using Owners name:



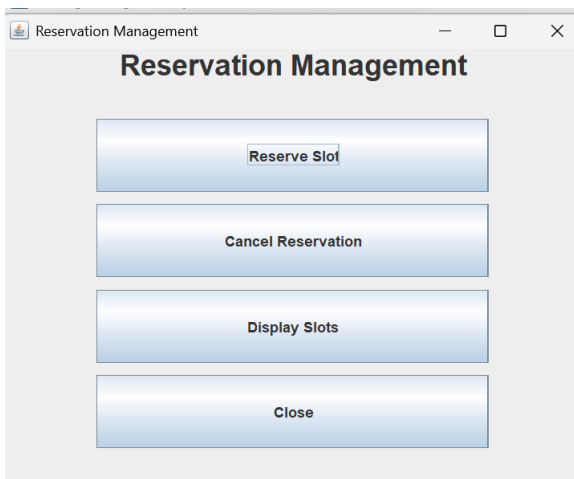
iii)To Display parked slots



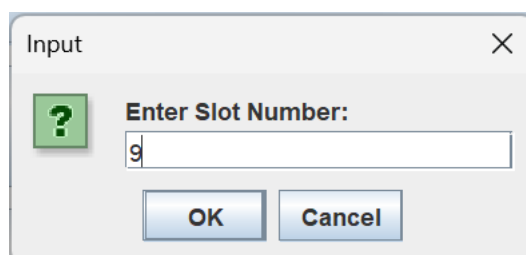
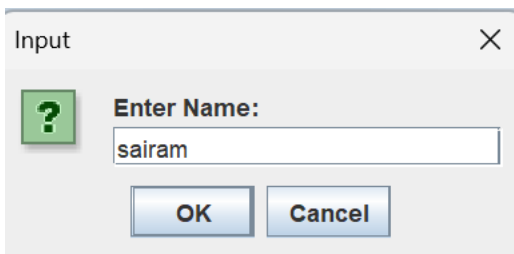
Command Prompt displaying data

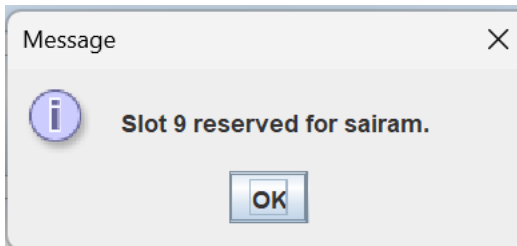
```
C:\Users\girij\Downloads\ParkingManagementSystem>java main.ParkingSystem
Parking data saved successfully.
Vehicle parked in Slot 1
Vehicle is already parked.
Parking data saved successfully.
Vehicle parked in Slot 2
Parking data saved successfully.
Vehicle parked in Slot 3
Parking data saved successfully.
Vehicle parked in Slot 4
Parking data saved successfully.
Vehicle removed from Slot 2
Parking data saved successfully.
Vehicle removed from Slot 1
Parking data saved successfully.
Vehicle parked in Slot 1
Parking data saved successfully.
Vehicle parked in Slot 2
```

5. Reservation Management



a) To reserve a slot:

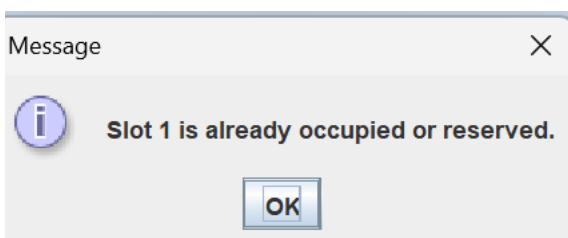




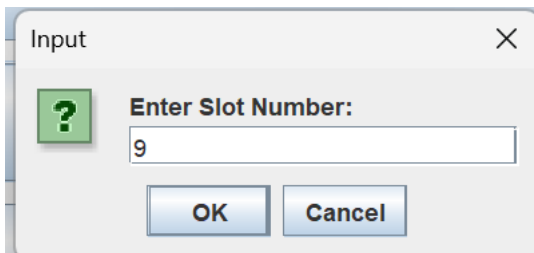
b) To display Reserved slots:



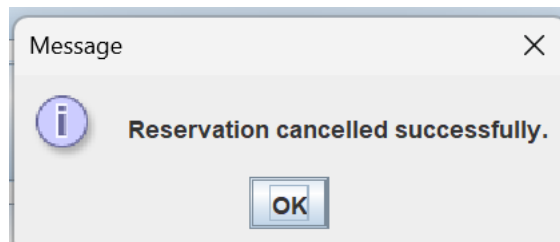
Trying to Reserve an already occupied slot



c) To Cancel a reserved slot:

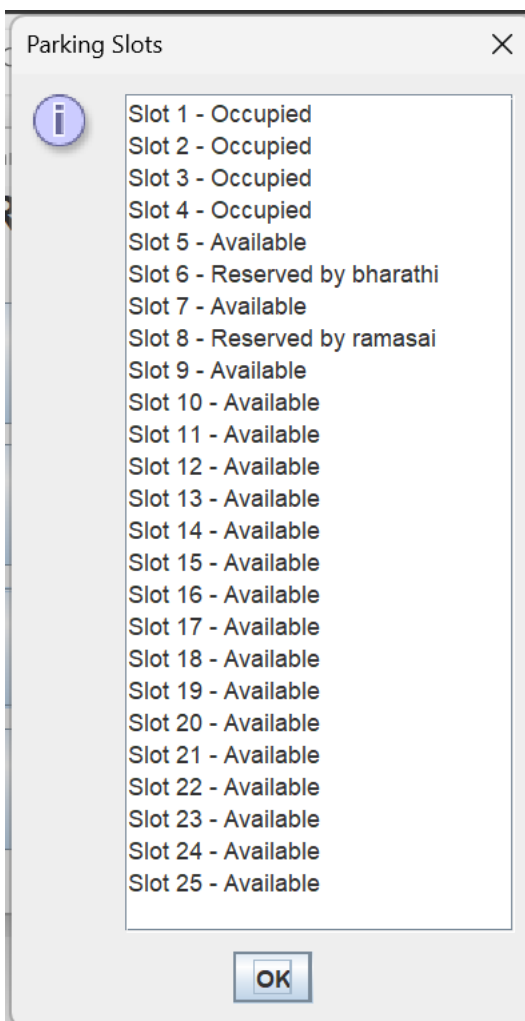


An input dialog box titled "Input" with a close button (X) in the top right corner. It contains a green question mark icon on the left. The main text is "Enter Slot Number:". Below this is a text input field containing the number "9". At the bottom are two buttons: "OK" and "Cancel".



A message dialog box titled "Message" with a close button (X) in the top right corner. It contains a blue information icon (i) on the left. The main text is "Reservation cancelled successfully.". At the bottom is an "OK" button.

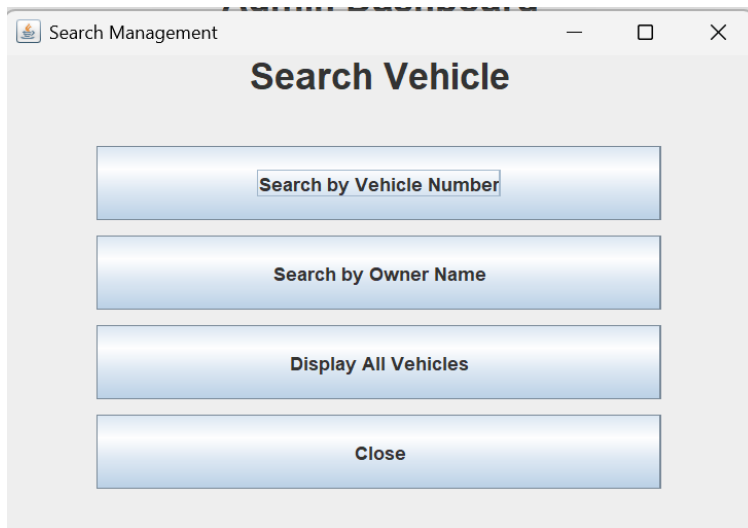
Display after cancellation of reservation:



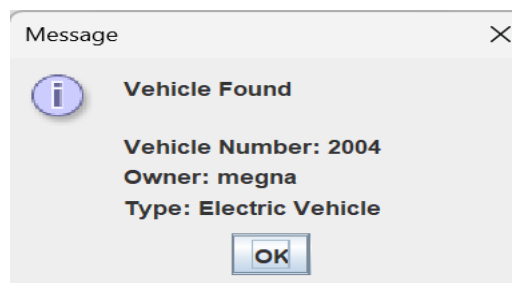
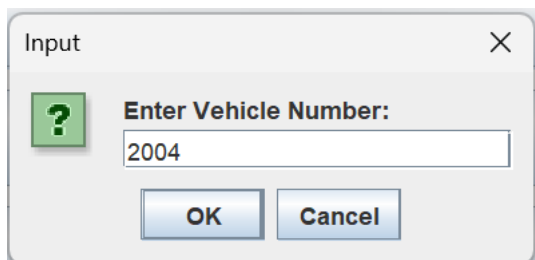
A "Parking Slots" dialog box with a close button (X) in the top right corner. It features a blue information icon (i) on the left. The main area is a list of 25 slots, each with its status. At the bottom is an "OK" button.

Slot Number	Status
Slot 1	Occupied
Slot 2	Occupied
Slot 3	Occupied
Slot 4	Occupied
Slot 5	Available
Slot 6	Reserved by bharathi
Slot 7	Available
Slot 8	Reserved by ramasai
Slot 9	Available
Slot 10	Available
Slot 11	Available
Slot 12	Available
Slot 13	Available
Slot 14	Available
Slot 15	Available
Slot 16	Available
Slot 17	Available
Slot 18	Available
Slot 19	Available
Slot 20	Available
Slot 21	Available
Slot 22	Available
Slot 23	Available
Slot 24	Available
Slot 25	Available

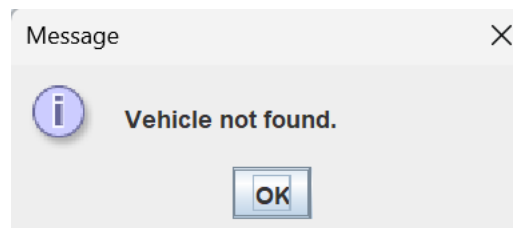
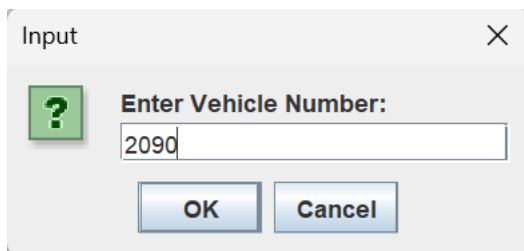
6. Search Management



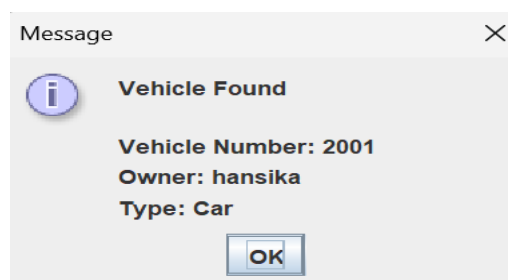
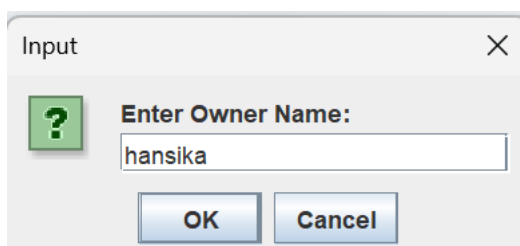
i) Search by Vehicle Number:



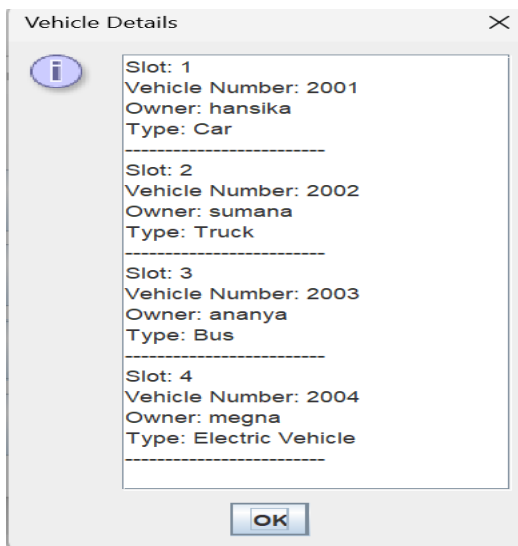
Case of Invalid Vehicle Number:



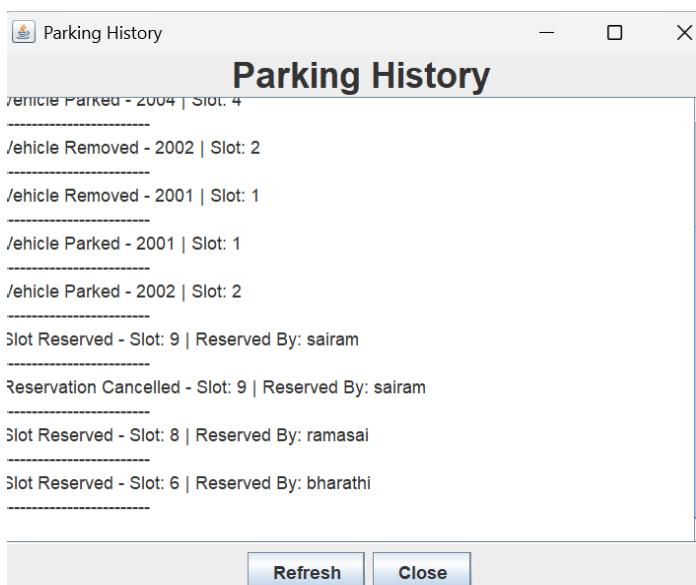
ii) Search by Owners Name:



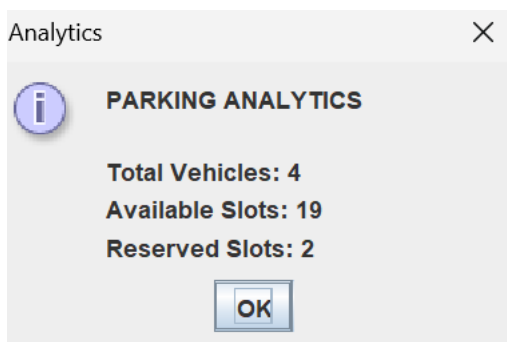
iii) Display all Vehicles



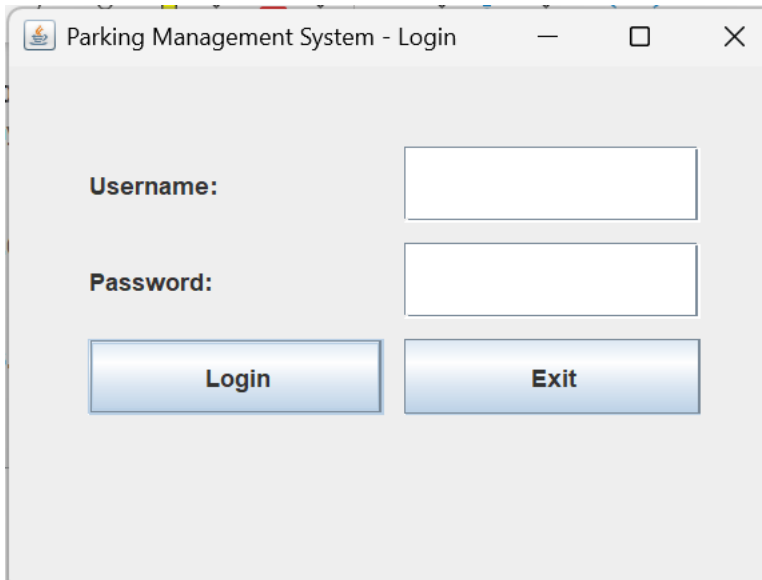
7. Parking History:



8. Parking Analytics:

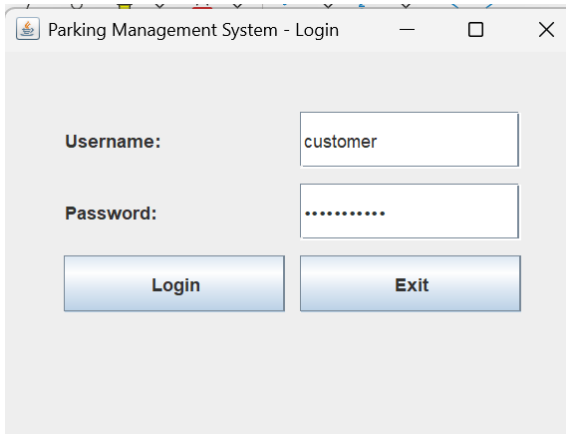


9. Logout (returns to login page):

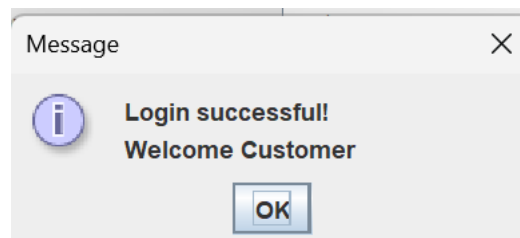


A screenshot of a Windows-style application window titled "Parking Management System - Login". The window has a light gray background and standard window controls (minimize, maximize, close) in the title bar. Inside the window, there are two labels: "Username:" and "Password:". To the right of each label is a white text input field. Below the input fields are two blue buttons with white text: "Login" and "Exit".

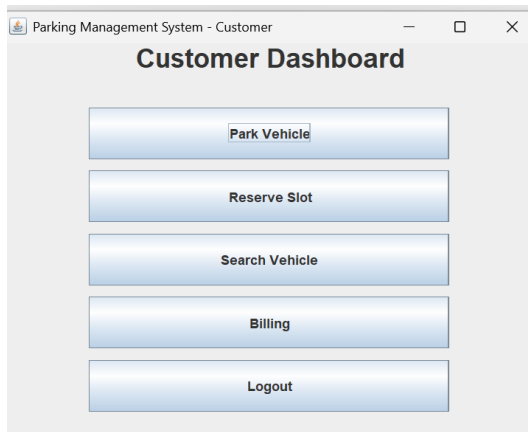
10. CUSTOMER LOGIN:



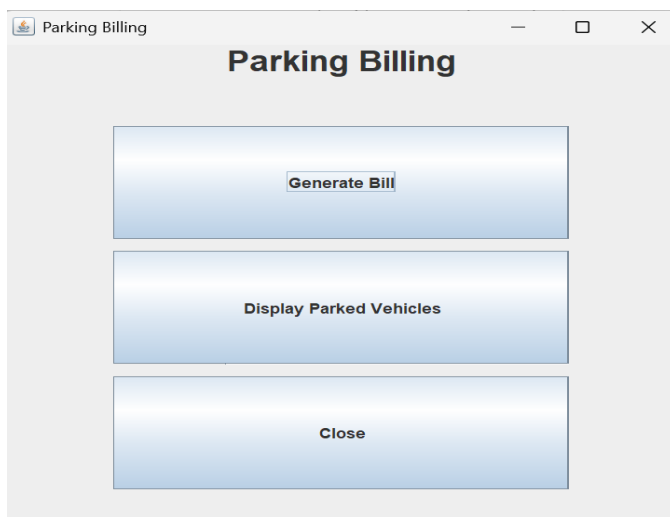
A screenshot of the same "Parking Management System - Login" window. The "Username:" input field now contains the text "customer". The "Password:" input field contains a series of dots, indicating a masked password. The "Login" and "Exit" buttons remain at the bottom.



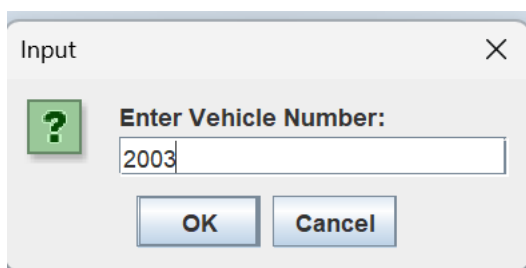
CUSTOMER DASHBOARD:



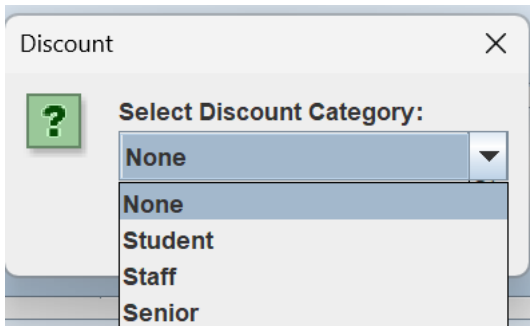
11. PARKING BILLING:



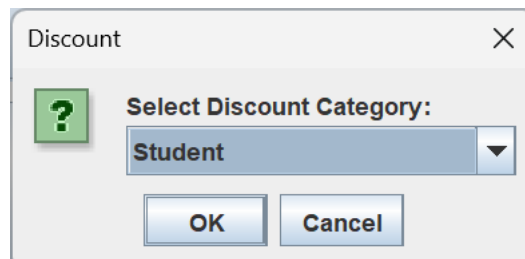
i) Generate Bill



Discount Category:

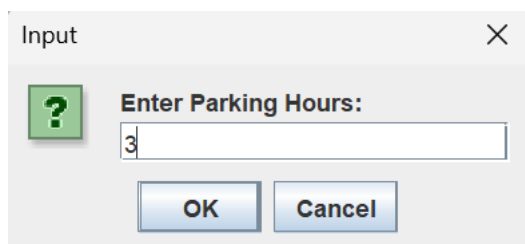


A dialog box titled "Discount" with a close button (X) in the top right corner. It contains a green square icon with a white question mark. Below the icon is the text "Select Discount Category:". A dropdown menu is open, showing the following options: "None", "None", "Student", "Staff", and "Senior".



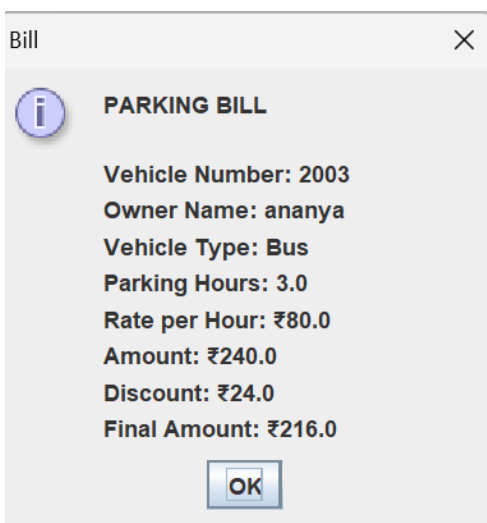
A dialog box titled "Discount" with a close button (X) in the top right corner. It contains a green square icon with a white question mark. Below the icon is the text "Select Discount Category:". A dropdown menu is open, showing the option "Student". Below the dropdown menu are two buttons: "OK" and "Cancel".

Number of Parking Hours:



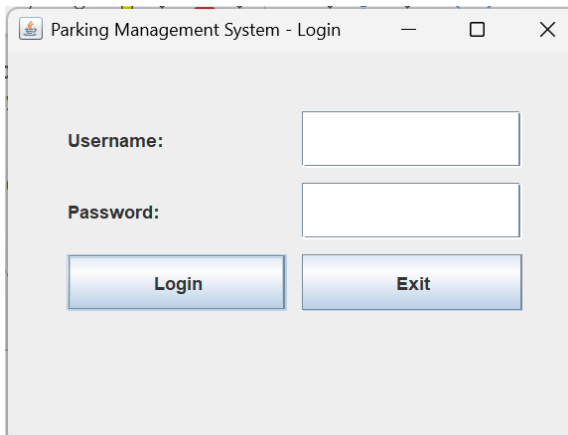
An input dialog box titled "Input" with a close button (X) in the top right corner. It contains a green square icon with a white question mark. Below the icon is the text "Enter Parking Hours:". A text input field contains the number "3". Below the input field are two buttons: "OK" and "Cancel".

Final Bill:



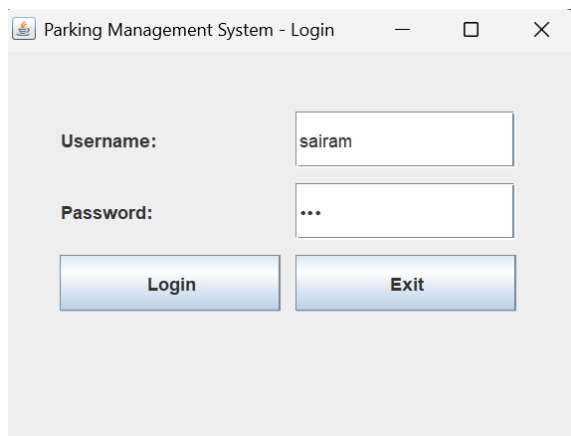
A dialog box titled "Bill" with a close button (X) in the top right corner. It contains a purple circular icon with a white 'i'. Below the icon is the text "PARKING BILL". The following details are listed:
Vehicle Number: 2003
Owner Name: ananya
Vehicle Type: Bus
Parking Hours: 3.0
Rate per Hour: ₹80.0
Amount: ₹240.0
Discount: ₹24.0
Final Amount: ₹216.0
At the bottom is an "OK" button.

12. After Logging out as Customer (returns to login page) :

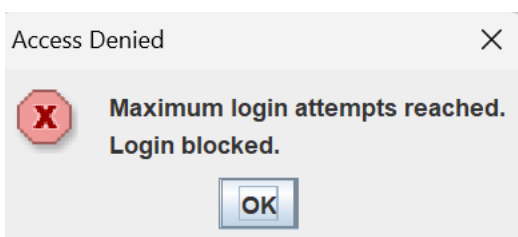
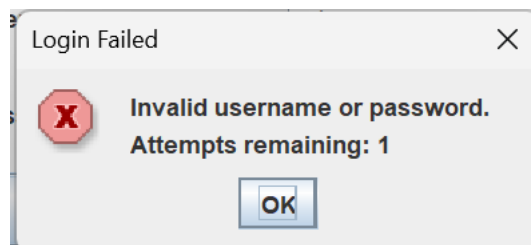
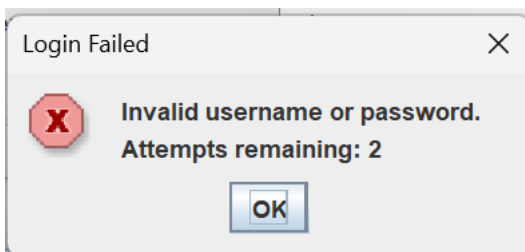


The screenshot shows a window titled "Parking Management System - Login". It contains two input fields: "Username:" and "Password:". Below the fields are two buttons: "Login" and "Exit".

13. Login Attempt Failure:



The screenshot shows the same "Parking Management System - Login" window. The "Username:" field now contains the text "sairam". The "Password:" field contains three dots "...". The "Login" and "Exit" buttons are still present.



Application closes after Denial of Access

XII. ADVANTAGES:

- Easy to use and understand
- Efficient management of vehicles and parking slots
- Reduces manual errors in parking and billing activities
- Provides parking slot reservation and cancellation
- Supports vehicle search and specific vehicle removal
- Maintains parking history and records
- Provides role-based access for Admin and Customer
- Modular and scalable design using Java OOP concepts
- Provides a user-friendly graphical interface using Java Swing

XIII. LIMITATIONS:

- Uses file handling instead of a database for data storage
- Parking duration is entered manually for billing
- History records are maintained during the application session
- Analytics provides basic parking information
- Security features are limited to basic username and password authentication

XIV. CONCLUSION:

The Smart Parking Management System demonstrates the practical application of Java Object-Oriented Programming concepts to a real-world parking scenario. The system provides organized management of vehicles, parking slots, reservations, billing, searching, history, and analytics through a user-friendly Java Swing interface. The modular design improves code organization, readability, and maintainability. Overall, the project provides a practical implementation of Java concepts such as classes and objects, inheritance, abstraction, encapsulation, interfaces, collections, exception

handling, file handling, and GUI programming, while providing a foundation for further development of a more advanced parking management system.

-----end of report-----